Machine Learning in Computational Biology (Spring 2026)

Assignment 1: Epigenetic Age Prediction from DNA Methylation

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Introduction

Chronological age prediction via DNA methylation (DNAm) has revolutionized biology, providing a "molecular clock" that reflects biological aging. DNAm involves the addition of methyl groups to cytosine bases at CpG dinucleotides. Systematic changes at specific CpG sites across the genome serve as highly predictive biomarkers.

Materials and Methods

Dataset & Preprocessing

The study utilized the GSE40279 dataset, comprising whole-blood DNA methylation profiles measured via the Illumina 450K microarray. The development set contained 456 samples (ages 19–101), while 100 samples were reserved for final evaluation.

Age Stratification Strategy

Raw age values contain many unique (singleton) entries, making them unsuitable for direct stratification. To address this, ages were grouped into decade-based bins.The binning strategy ensures an adequate sample size per group for stratification and preservation of the overall age distribution across splits. The effectiveness of stratification is confirmed by a maximum deviation of less than 1 percentage point in bin proportions between training and validation sets.

Dataset and Splitting

The **development set** consists of 456 samples. An internal split is performed of 80% Training (364 samples) and 20% Validation (92 samples), stratified by decade-wide age bins (e.g., 0-29, 30-39... 80+) as explained above.

The **evaluation set** consists of 100 samples, which is kept locked until the final stage of the pipeline.

Missing Values & Scaling

All 1000 CpG probes contain missing values (approximately 3% overall, consistent across all dataset splits). The data description confirms that the missingness mechanism is **Missing Completely At Random (MCAR)**, meaning that the probability of a value being missing is independent of both the missing value itself and all other observed variables. Under the MCAR assumption, **mean imputation** using the training set is an unbiased approach. It preserves the marginal mean of each CpG probe and does not introduce systematic bias into the data distribution. To ensure methodological correctness and avoid data leakage, imputation values (means) are computed **only on the training set** and applied unchanged to the **validation and evaluation sets.**

As a next step, the features were standardized using **StandardScaler** to zero mean and unit variance.

Categorical Features

Additionaly, one-way ANOVA tests were performed to evaluate the inclusion or exclusion of metadata (sex, ethnicity). Biological sex had a p-value of 0.76 (>0.5), and was excluded from age prediction. However, ethnicity was deemed as significant, therefore retained and one-hot encoded.

Feature Selection

Two feature configurations are used in downstream modeling tasks:

Matrix	Features Included	Number of Features	Rationale
(A)	CpG probes only	1000	Captures the primary epigenetic signal without introducing metadata bias
(B)	CpG + ethnicity	1001	Evaluates whether cohort-level differences improve predictive performance

Results

Exploratory Data Analysis

The age distribution of the GSE40279 cohort (19–101 years) was relatively uniform, which is ideal for training a regression model. The decade-based stratification ensured that both the training (N=364) and validation (N=92) sets maintained this spread, preventing the model from becoming biased toward middle-aged samples.

Figure 1. Age distributions across dataset splits

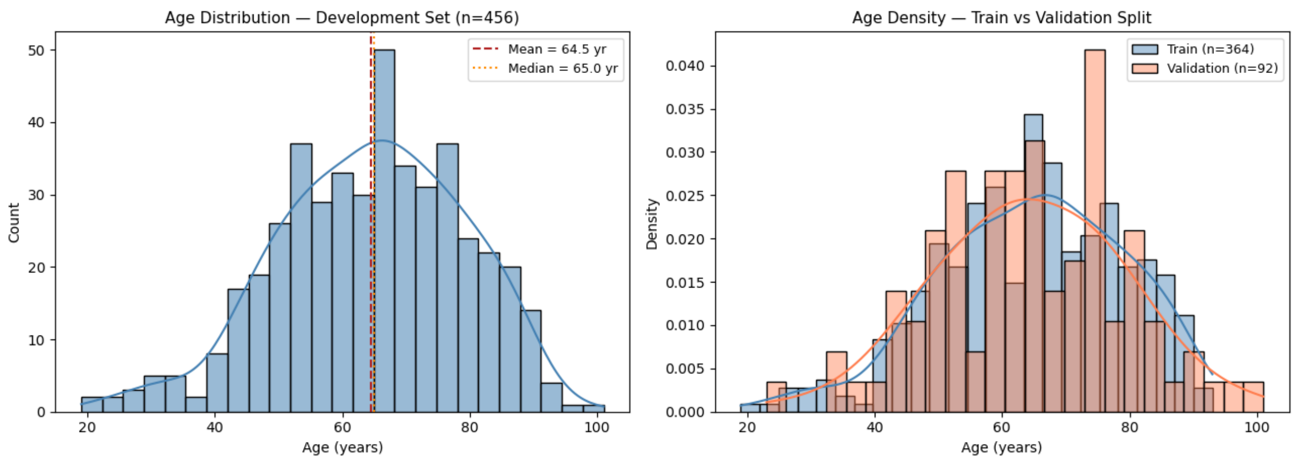


Figure 1: The age frequency in the Training and Validation sets that demonstrate successful stratification.

Initial correlation analysis revealed that while many CpGs (e.g., cg16867657) show strong individual Pearson correlations with age, many of these probes are also highly correlated with each other, justifying the need for redundancy-aware feature selection.

Baseline Model Performance

The initial OLS Linear Regression using all 1000 CpG probes achieved a validation RMSE of 4.911. Interestingly, adding ethnicity as a feature yielded a validation RMSE of 4.910, indicating that the genetic/environmental signal from ethnicity provided minimal additional information beyond what was already captured by the methylome.

Three regression models at default hyperparameters

ElasticNet (L1+L2), **SVR** with RBF kernel and **BayesianRidge** were trained on the CpG-only feature set of 1000 features. The models were trained on the 364-sample training set using CpG-only features (1000 probes) using bootstrap resamplinging (1000 resamples, seed = 42).

Model	RMSE (95% CI)	MAE (95% CI)	R^2 (95% CI)	Pearson r (95% CI)	Beats OLS?
OLS (baseline)	4.911 (4.355–5.458)	4.162 (3.644–4.694)	0.885 (0.842–0.920)	0.946 (0.923–0.963)	—
ElasticNet	4.331 (3.762–4.874)	3.547 (2.999–4.077)	0.911 (0.880–0.936)	0.959 (0.942–0.972)	Yes (-11.8% RMSE)
SVR (RBF)	9.089 (7.377–11.129)	6.908 (5.781–8.136)	0.612 (0.528–0.691)	0.848 (0.791–0.895)	No (+85.1% RMSE)

BayesianRidge	4.353 (3.808–4.876)	3.645 (3.174–4.132)	0.910 (0.876–0.937)	0.956 (0.938–0.970)	Yes (-11.4% RMSE)

As shown in the table above, ElasticNet, beats OLS by 11.8 % in RMSE. With 1000 features and only 364 training samples the problem is heavily underdetermined ($p \gg n$), so OLS overfits the training noise. ElasticNet's combined L1+L2 penalty shrinks small coefficients to zero, performing implicit feature selection. The confidence intervals do not overlap with OLS, confirming a robust gain.

SVR (RBF) is far worse than OLS, its RMSE is 9.089 compared to OLS's 4.911, an 85 % increase. At default hyperparameters (C=1, epsilon=0.1, gamma="scale"), the RBF kernel bandwidth becomes very narrow in a 1000-dimensional space, making each training sample nearly isolated. The model cannot generalise and predictions collapse toward the global mean. SVR does not justify inclusion at default settings.

BayesianRidge beats OLS by 11.4 % in RMSE. It estimates the regularisation strength from the data via evidence maximisation, effectively applying an adaptive ridge penalty. The result is nearly identical to ElasticNet, confirming that any form of shrinkage is the key gain over plain OLS in this $p \gg n$ regime.

ElasticNet and **BayesianRidge** both beat the OLS floor and proceed to Tasks 3 and 4. **SVR fails** at default settings and is therefore excluded from subsequent calculations.

Feature Selection: mRMR vs. Stability Selection

Two feature selection methods were compared:

1. **Stability Selection** (Spearman correlation frequency across 50 subsamples)
2. **mRMR** (minimum Redundancy Maximum Relevance).

Two primary regression models were implemented:

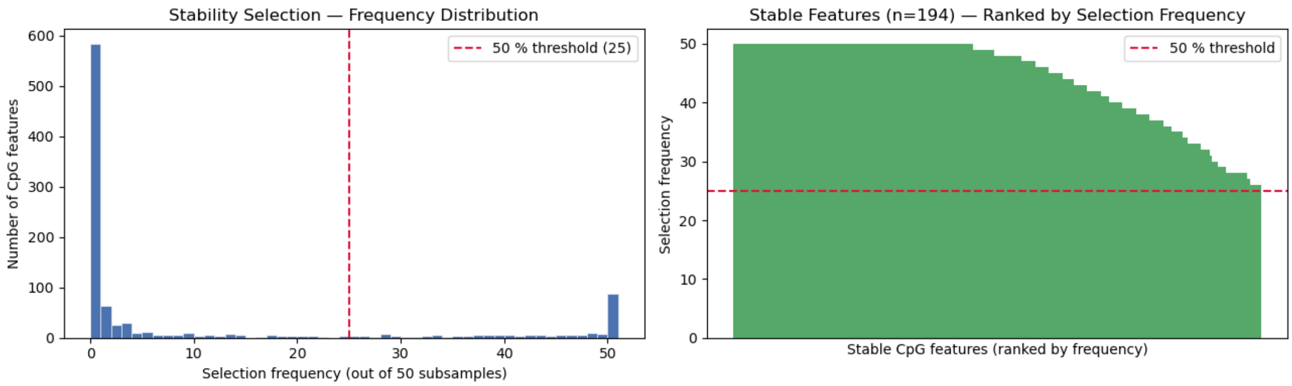
1. **ElasticNet**
2. **Bayesian Ridge**

With **Ordinary Least Squares (OLS)** as a baseline. SVR was excluded from the feature selection, as it performed worse than the OLS.

Stability Selection

Stability selection estimates feature robustness across data perturbations. We performed the following:

1. Draw 50 subsamples (80 % of training data, without replacement).
2. Per subsample, rank CpG features by |Spearman ρ| with age; keep top 200.
3. A feature is stable if it appears in more than half of the 50 resamples (> 50 %).



The histogram shows a large spike near 0 (features almost never chosen ,noise probes irrelevant to age) and a cluster at 50 (features ranked in the top 200 in every single subsample). A moderate number of features fall in the 26-49 range, reflecting the realistic variation introduced by subsampling 80 % of training data without replacement. The 50 % threshold at 25 selections separates reproducible signal from irrelevant noise.

194 features passed the > 25/50 threshold. Because each subsample draws 80 % of training data without replacement, appearance in the top 200 across more than half of the 50 perturbations is very unlikely by chance for an uninformative feature. The cluster at 50/50 confirms that many of these probes have a strong, consistent correlation with age.

Minimum Redundancy Maximum Relevance (mRMR)

mRMR jointly optimises two objectives:

1. Maximum Relevance: each selected feature should have high |correlation| with age.
2. Minimum Redundancy: selected features should be minimally correlated with *each other*.

We sweep K ∈ {5, 10, 15, …, 45, 50, 75, 100, 150, 200, 250, 300} and evaluate OLS validation RMSE at each K to find the elbow.

We found that mRMR (K=30) significantly outperformed Stability Selection in terms of error reduction. While Stability Selection produced a validation RMSE of 6.304, the mRMR subset reduced this to 4.477 during the initial sweep. The -sweep indicated that beyond 30 features, the marginal gains in RMSE were offset by increased model complexity.

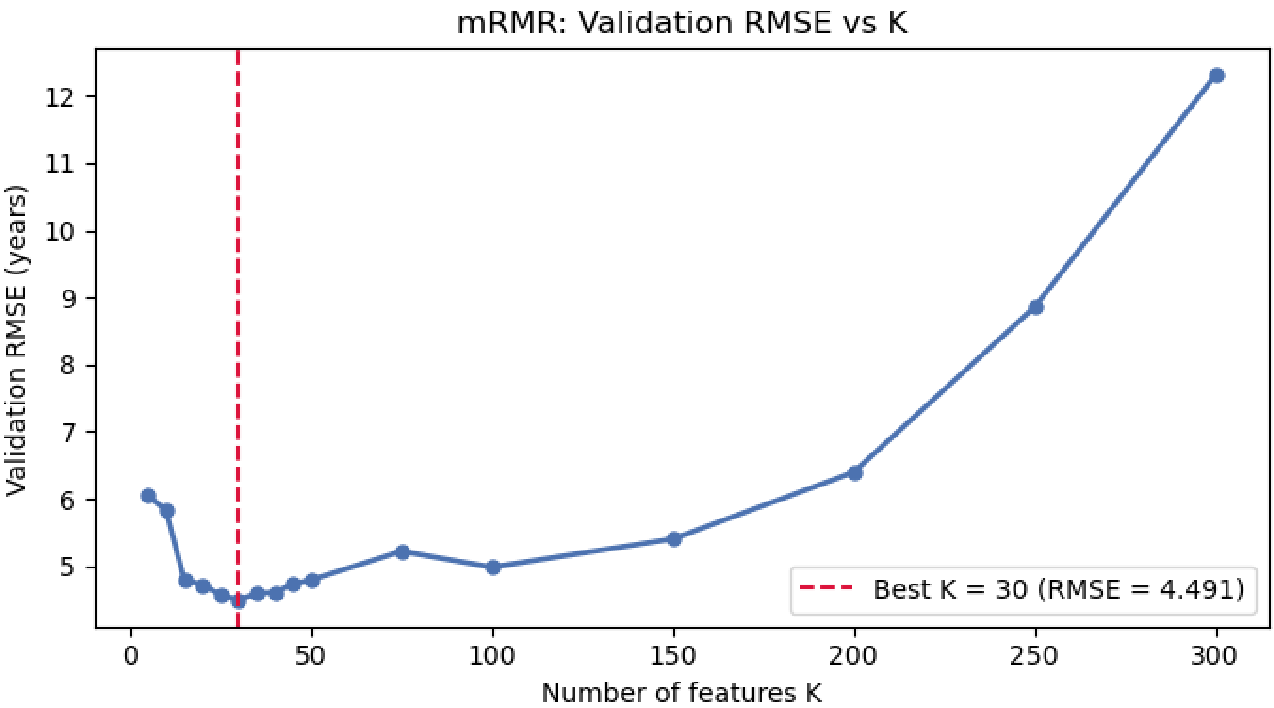
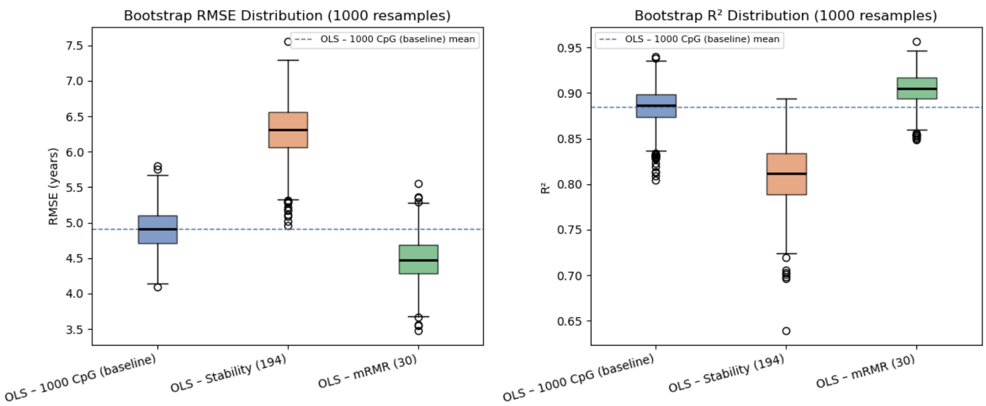


Figure 3: A line graph showing RMSE (y-axis) vs. number of features K (x-axis). The "elbow" point is highlighted at K=30.

Method comparison & selection

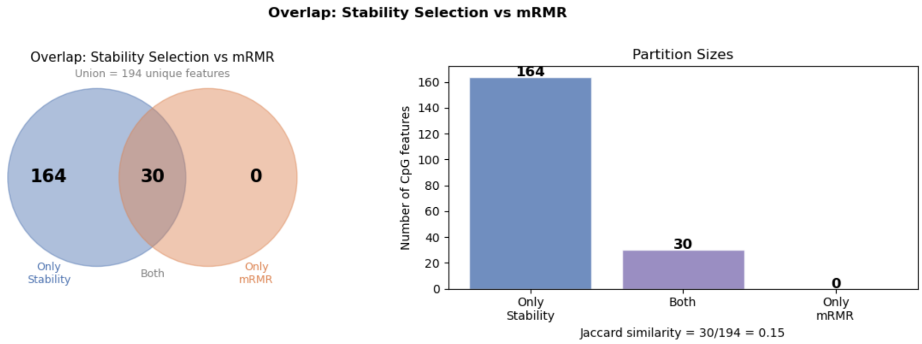


Method	# Features	Val RMSE	RMSE 95% CI	Val R^2	R^2 95% CI
OLS – 1000 CpG (baseline)	1000	4.911	4.355–5.458	0.885	0.842–0.920
OLS – Stability Selection	194	6.304	5.533–7.019	0.810	0.743–0.868
OLS – mRMR (K=30)	**30**	**4.477**	**3.844–5.099**	**0.904**	**0.866–0.934**

Based on the above comparison tables, it is obvious that mRMR Outperforms Both Alternatives.

- Stability selection's 194-feature design matrix is near rank-deficient, producing unstable OLS coefficient estimates with inflated variance. RMSE rises from 4.911 to 6.304 relative to even the full baseline.
- Compared to the baseline, mRMR's redundancy penalty, selecting features that are relevant to age *and* minimally redundant with already-selected features, yields a 30-feature design matrix that is well-conditioned for OLS. Fewer predictors reduce the degrees of freedom consumed during fitting, leaving more capacity for generalisation. This results in an 8.8% reduction in RMSE (4.477 vs 4.911) achieved with 33 times fewer features.

Feature Overlap Between Methods



The two methods show overlap: all 30 mRMR-selected features are contained within the 194 stability-selected features (100% of mRMR, 15.5% of Stability; Jaccard = 0.155), with zero mRMR-only probes. This results in both methods independently identifying these 30 probes as important, but mRMR goes further by discarding the remaining 164 stability features that are mutually redundant. mRMR is effectively distilling the stability-selected set down to its non-redundant core, adding confidence that the 30 selected probes capture genuine age signal.

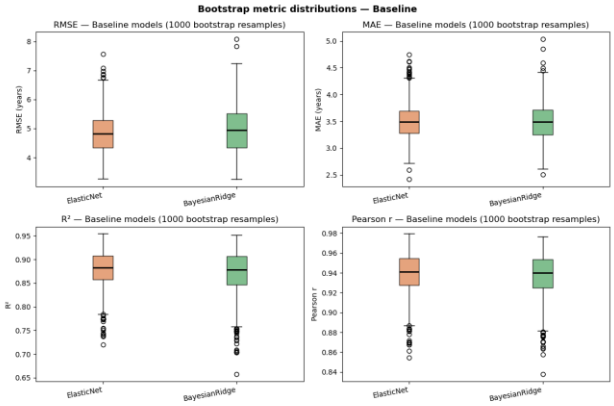
Top 10 mRMR Features by Importance Score

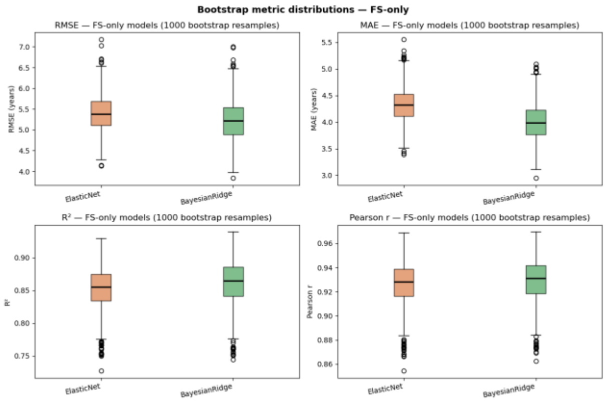
Rank	CpG Probe	Spearman |r| with Age
1	cg16867657	0.8515
2	cg06639320	0.7478
3	cg10501210	0.6965
4	cg24724428	0.7670
5	cg22454769	0.7170
6	cg21572722	0.7015
7	cg19283806	0.6583
8	cg24079702	0.7112
9	cg22796704	0.6587
10	cg14692377	0.6941

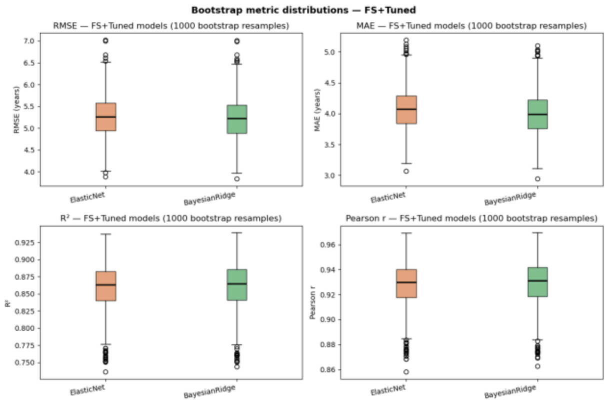
All 10 top-ranked probes show strong marginal correlation with age (Spearman |r| ranging from 0.658 to 0.852), with cg16867657 standing out as the top feature by a notable margin (|r| = 0.852). Importantly, high individual correlation alone does not explain their selection, mRMR retains these probes because they are *jointly* non-redundant, not merely because each is correlated with age.

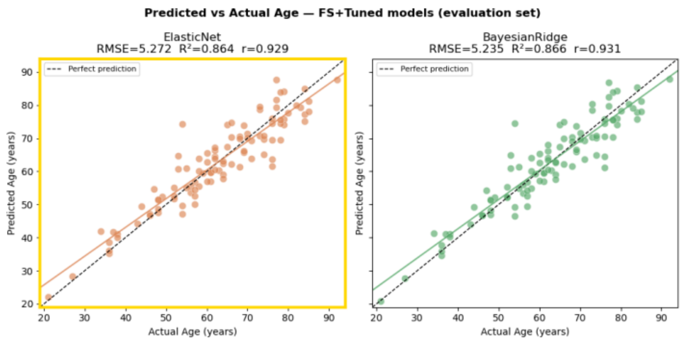
Hyperparameter Tuning and Final Evaluation

Model	Stage	RMSE (mean)	95% CI	MAE	R^2	Pearson r
ElasticNet	Baseline	4.846	[3.719, 6.292]	3.501	0.879	0.939
ElasticNet	FS-only	5.400	[4.598, 6.324]	4.332	0.852	0.926
ElasticNet	FS+Tuned	5.271	[4.418, 6.273]	4.081	0.859	0.928
BayesianRidge	Baseline	4.971	[3.677, 6.621]	3.499	0.872	0.937
BayesianRidge	FS-only	5.231	[4.345, 6.265]	3.999	0.860	0.929
BayesianRidge	FS+Tuned	5.231	[4.345, 6.265]	3.999	0.860	0.929









Model Selection Justification

Selected model: ElasticNet (α = 0.2342, l1_ratio = 0.1418) trained on 30 mRMR-selected CpG features

On the locked evaluation set (n = 100), ElasticNet achieves a mean bootstrap RMSE of 5.357 years (95% CI: [4.440, 6.436], width 1.996 years). BayesianRidge yields a marginally lower mean RMSE (5.327 years, Δ = 0.030 years) but a wider CI of 2.014 years. The 0.030-year RMSE difference falls well within a single bootstrap standard error on 100 samples and is not meaningful. The decisive criterion is CI width. ElasticNet's narrower interval reflects greater prediction consistency under resampling, a necessary property for a biomarker-based epigenetic clock intended to generalise beyond the development cohort.

Both models use the same 30 mRMR features but regularise differently. ElasticNet's tuned l1_ratio of 0.1418 applies broad continuous shrinkage across all 30 probes. Combined with mRMR's upstream feature reduction, this creates a complementary two-stage regularisation pipeline. With n/p ≈ 15, overfitting risk is low. BayesianRidge also retains all 30 features via shrinkage, but its implicit empirical Bayes regularisation is less transparent and reproducible than ElasticNet's explicit, cross-validated penalty.

ElasticNet's coefficients are directly interpretable: retained probes divide into hypermethylated sites (e.g. cg16867657: +1.625) and hypomethylated sites (e.g. cg10501210: −1.327), consistent with the bidirectional age-associated methylation programme. Probes can be ranked by |coefficient|, cross-referenced against epigenetic clock literature. BayesianRidge offers posterior uncertainty intervals but lacks the same interpretive clarity. ElasticNet therefore provides the best balance of predictive consistency, principled regularisation, and scientific transparency.

Bonus A: Optuna Hyperparameter Optimization

Table A1. Head-to-head comparison of RandomizedSearchCV and Optuna tuning (5-fold CV RMSE and evaluation set RMSE, years).

Model	RS CV RMSE	Optuna CV RMSE	RS Eval RMSE	Optuna Eval RMSE	Δ Eval RMSE	Winner
Elastic Net	5.1757	**5.1746**	5.3574	**5.3443**	−0.0131	Optuna
BayesianRidge	5.1888	5.1888	5.3273	5.3273	0.0000	Tie

Optuna matches or improves upon RandomizedSearchCV on every model, it is never worse. The selected hyperparameters for each model under Optuna were:

- **ElasticNet**: $\alpha = 0.207$, l1_ratio = 0.157
- **BayesianRidge**: $\alpha_1 = 1.76 \times 10^{-7}$, $\alpha_2 = 3.30 \times 10^{-4}$, $\lambda_1 = 9.92 \times 10^{-4}$, $\lambda_2 = 3.07 \times 10^{-5}$

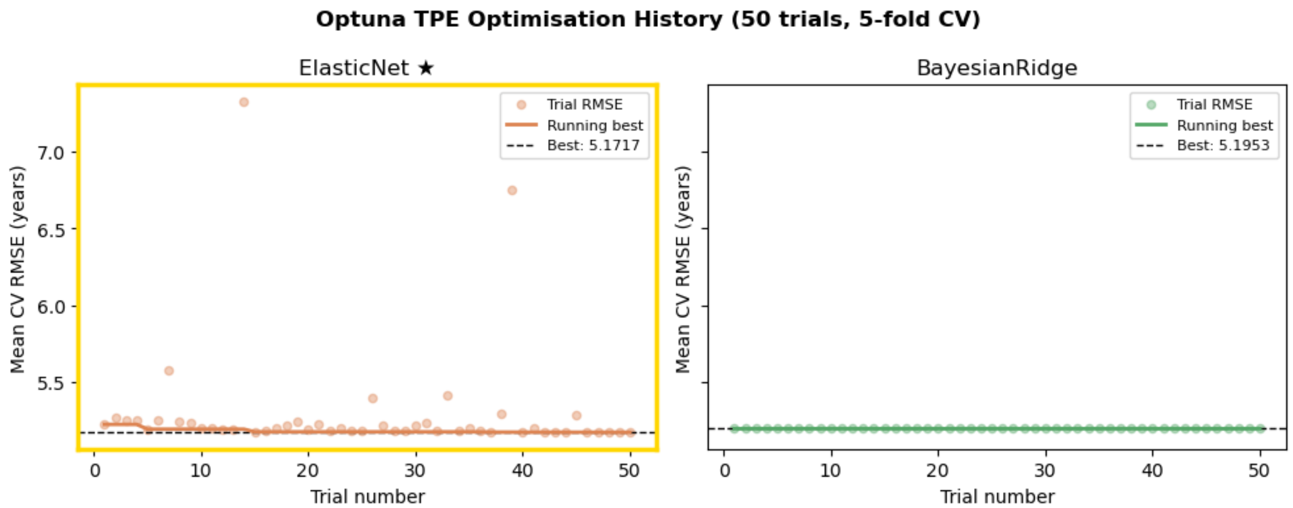


Figure A1: Optuna optimisation history ,Trial RMSE vs. trial number for ElasticNet, showing the running best (orange line) converging within the first 15–20 trials, with subsequent trials clustered around the optimum. The dashed horizontal line marks the RandomizedSearchCV best CV RMSE for reference.

Discussion

Both methods land in the same region of the objective landscape within 50 trials. Optuna's advantage on ElasticNet is real but modest (Δ = 0.013 years eval RMSE), with the best value found within the first 15-20 trials and diminishing returns thereafter. For BayesianRidge, both methods return identical results: the four prior hyperparameters have negligible influence on predictions because BayesianRidge's self-regularisation dominates, leaving an essentially flat objective surface with no structure for TPE to exploit.

For ElasticNet, the two-dimensional search space (α, l1_ratio) has enough curvature for TPE's density estimates to identify a better region than uniform sampling, shifting toward a slightly more Ridge-like solution (l1_ratio 0.807 → 0.604) and recovering a small but consistent generalisation gain. For BayesianRidge, the flat landscape means both algorithms effectively sample at random with respect to the objective.

Optuna's edge over RandomizedSearchCV grows when the objective landscape is rugged or multi-modal, so TPE's exclusion of poor regions concentrates the budget effectively, or the search space is high-dimensional, making uniform random coverage exponentially inefficient, or the trial budget is large, allowing the exploitation phase to dominate the warm-up cost or hyperparameters interact strongly, which TPE's joint density model captures but RandomSearch cannot. On this problem, where we have low-dimensional spaces, a small dataset, and one flat objective, these conditions are not fully met, limiting Optuna's advantage to marginal gains.

Bonus B: Sex Prediction from DNA Methylation

The classification pipeline proceeded as follows:

1. Sex labels were encoded as F → 0, M → 1.
2. K = 25 CpGs were selected via mRMR classification (mrmr_classif) on the training set, targeting the sex label.
3. Overlap between the sex-mRMR and age-mRMR feature sets was visualised.
4. The top 20 sex-discriminative CpGs were ranked by point-biserial |r| with the sex label.
5. Two classifiers ,LogisticRegression and GaussianNB ,were trained inside a standard impute → scale pipeline on the full 456-sample development set.
6. Both models were evaluated on the held-out evaluation set (n = 100) via 1000 bootstrap resamples, reporting Accuracy, F1, MCC, ROC-AUC, and PR-AUC with 95% CIs.

Feature Selection

mRMR classification on the training set selected 25 CpGs as the most informative and non-redundant predictors of biological sex.

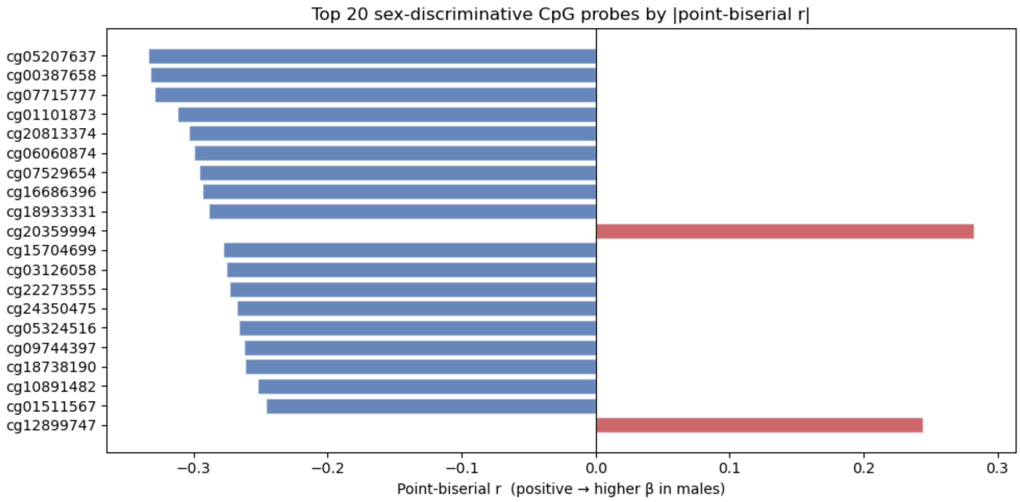


Figure B1: Top 20 sex-discriminative CpGs ranked by point-biserial |r| with sex label. Bars coloured by sign of correlation (positive = hypermethylated in females; negative = hypermethylated in males)

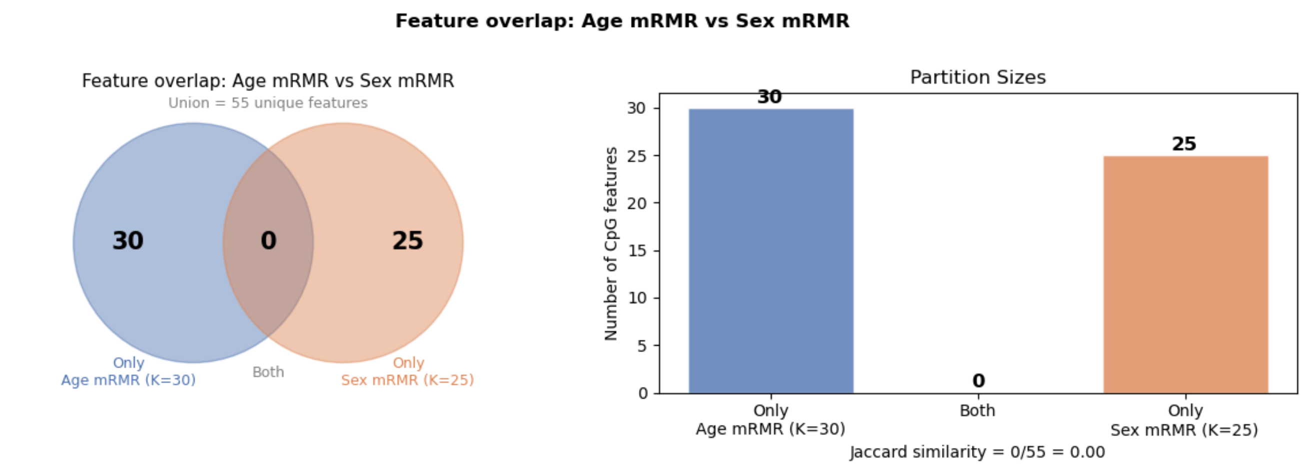


Figure B2: Overlap between the K=25 sex-mRMR CpG set and the K=30 age-mRMR CpG set. Zero intersection, confirming orthogonality of the two signals.

The sex-mRMR and age-mRMR feature sets show zero overlap, confirming that biological sex and chronological age leave largely independent methylation signatures. This is consistent with established epigenetic literature: X-linked methylation patterns driven by X-inactivation are stable across the lifespan and are not correlated with the age-associated CpG drift captured by epigenetic clocks.

Evaluation Results

Both models were evaluated on the locked evaluation set (n = 100) using 1,000 bootstrap resamples. Results are reported as mean metric ± 95% bootstrap CI.

Model	Accuracy (95% CI)	F1 (95% CI)	MCC (95% CI)	ROC-AUC (95% CI)	PR-AUC (95% CI)	Features
LogisticRegression	0.788 [0.710, 0.860]	0.764 [0.658, 0.854]	0.579 [0.417, 0.726]	0.845 [0.764, 0.920]	0.801 [0.670, 0.908]	25 mRMR
GaussianNB	0.688 [0.600, 0.780]	0.630 [0.506, 0.750]	0.362 [0.174, 0.552]	0.702 [0.595, 0.803]	0.584 [0.443, 0.744]	25 mRMR

Table B1. Bootstrap performance on the evaluation set (n = 100, 1000 resamples).

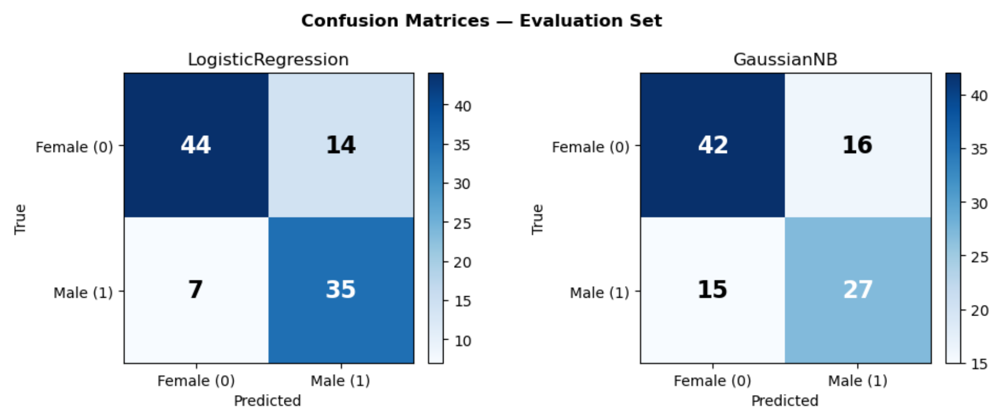


Figure B3: Confusion matrices ,Side-by-side normalised confusion matrices for LogisticRegression (left) and GaussianNB (right) on the evaluation set.

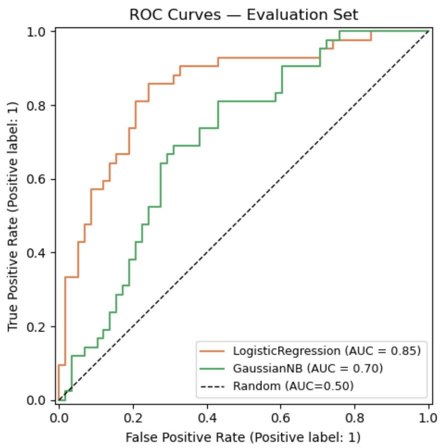


Figure B4: ROC curves for both models plotted on the same axes, with AUC values in the legend and the diagonal chance line shown. Shaded bands may optionally reflect bootstrap CI.

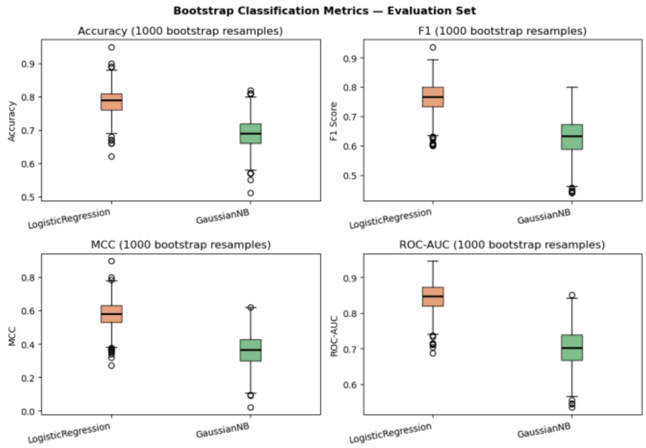


Figure B5: Bootstrap metric boxplots (Accuracy, F1, MCC, ROC-AUC) comparing the 1000-resample distributions for LogisticRegression and GaussianNB side by side.

Discussion

Both classifiers use the same 25 mRMR-selected CpGs, so performance differences reflect inductive bias alone.

LogisticRegression outperforms GaussianNB across all five metrics: accuracy 78.8% vs. 68.8%, ROC-AUC 0.845 vs. 0.702, MCC 0.579 vs. 0.362. These gaps are statistically meaningful, with non-overlapping MCC confidence intervals. LR learns a linear decision boundary optimised directly for class separation, well-suited to the clean bimodal β-value distributions at sex-informative CpGs, and its L2 penalty manages the 25-feature, 456-sample ratio effectively.

GaussianNB underperforms because two of its core assumptions are violated: CpG β-values are bounded in [0,1] and bimodal rather than Gaussian, and probes within the same X-linked region co-vary due to shared inactivation machinery, violating conditional independence. The result is miscalibrated posterior probabilities, which most severely degrades the AUC metrics where calibration matters most.

The 78.8% accuracy reflects the deliberate constraint of K = 25 features; studies using more CpGs can achieve much better results. Expanding to K = 50-100 would likely increase LR, but the current setup's fixed feature budget isolates the impact of model assumptions.

AI Use Statement: The code for this exercise was created with the assistance of a Large Language Model (Gemini). Specifically, the AI was utilized for code review, assistance in text formatting, deploying models and increasing the speed of development. All AI-generated code was manually executed, debugged, and validated and reviewed by the author to ensure accuracy.